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Experiment Number : 08

Experiment Title : Demonstration of Signal Transmission through Wireless Media (Wi-Fi)

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Experiment No. 08: Demonstration of Signal Transmission through Wireless Media (Wi-Fi)

Theory

1. What is Wireless Media? 📡

Wireless media refers to communication methods that transmit data without physical cables, using electromagnetic signals such as radio waves, microwaves, or infrared.

Instead of wires, devices communicate through the air using antennas and wireless technologies.

Examples:

- Wi-Fi
- Bluetooth
- Cellular networks (4G/5G)
- Satellite communication

Key advantages:

- Mobility and flexibility
- Easy installation
- Reduced wiring cost

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2. Define SSID (Service Set Identifier) and its purpose in a Wi-Fi network 📶

SSID (Service Set Identifier) is the name of a Wi-Fi network that helps devices identify and connect to the correct wireless network.

Purpose:

- Identifies a specific Wi-Fi network
- Allows users to select the correct network from a list
- Helps separate multiple networks in the same area
- Can be hidden for additional security

Example:
If your Wi-Fi name is “Home_Network_5G”, that name is the SSID.

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3. Difference between Wired LAN and Wireless LAN (WLAN) 🔌📡

Feature	Wired LAN	Wireless LAN (WLAN)
Connection Type	Uses cables (Ethernet)	Uses radio signals (Wi-Fi)
Mobility	Limited mobility	High mobility
Speed	Generally faster and stable	Can be slightly slower
Installation	Requires physical wiring	Easy to set up
Security	More secure	Needs encryption (password)
Cost	Higher installation cost	Lower installation cost

Example:

- Wired LAN → Office desktop connected via Ethernet cable
- WLAN → Laptop connected through Wi-Fi

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4. Significance of "Signal Strength" in wireless transmission 📶

Signal strength indicates how strong the wireless signal is between a transmitter (router) and receiver (device).

Importance:

- Strong signal → faster internet speed
- Weak signal → slow speed or disconnection
- Affected by distance, walls, and interference
- Determines reliability of the wireless connection

Example:
If you move far from the router, signal strength decreases, causing slower internet.

Hardware Components Required

- Wireless Router (WRT300N): The central device that broadcasts the wireless signal and manages the network.
- Laptop-PT (x2): End devices equipped with wireless network interface cards.

- Smartphone (x1): A mobile device with built-in Wi-Fi capability.

Hardware and Connection Setup

- Wireless Standard: 2.4 GHz 802.11n
- Network ID (DHCP): 192.168.0.0 (Default for WRT300N)
- SSID (Network Name): MyWirelessNetwork
- Authentication: WPA2-Personal

Procedure

Step 1: Physical Topology Design

- Place one **Wireless Router (WRT300N)** in the center.
- Place two **Laptops** and one **Smartphone** near the router.
- Naming: Rename Smartphone following the convention (e.g., S11_01).
- **Observation:** Note that the Smartphone connects automatically (default settings), but the Laptops do not.

Step 2: Hardware Identification (Laptop Modification)

- Click on a **Laptop** > **Physical** tab.
- **Task:** Turn off the Laptop power. Remove the Ethernet module (RJ45) and drag the **WPC300N** (Wireless Module) into the slot. Turn the power back on.
- **Observation:** A dashed line appears between the Laptop and Router, indicating a wireless link.

Step 3: Wireless Router Configuration

- Click the **Wireless Router** > **Config** tab > **Wireless**.
- Set the **SSID** to MyWirelessNetwork.
- Select **WPA2-PSK** and enter a Passphrase (e.g., TSEC@2026).
- **Observation:** The dashed lines disappear because the clients now need the correct security key to connect.

Step 4: Connecting Clients to the Wireless Signal

- Click **Laptop** > **Desktop** tab > **PC Wireless**.
- Go to the **Connect** tab. Click **Refresh**. Select MyWirelessNetwork and click **Connect**.
- Enter the Passphrase TSEC@2026
- ***Repeat this for the second Laptop and the Smartphone (under Config > Wireless)***.

Step 5: Logical Addressing (DHCP Verification)

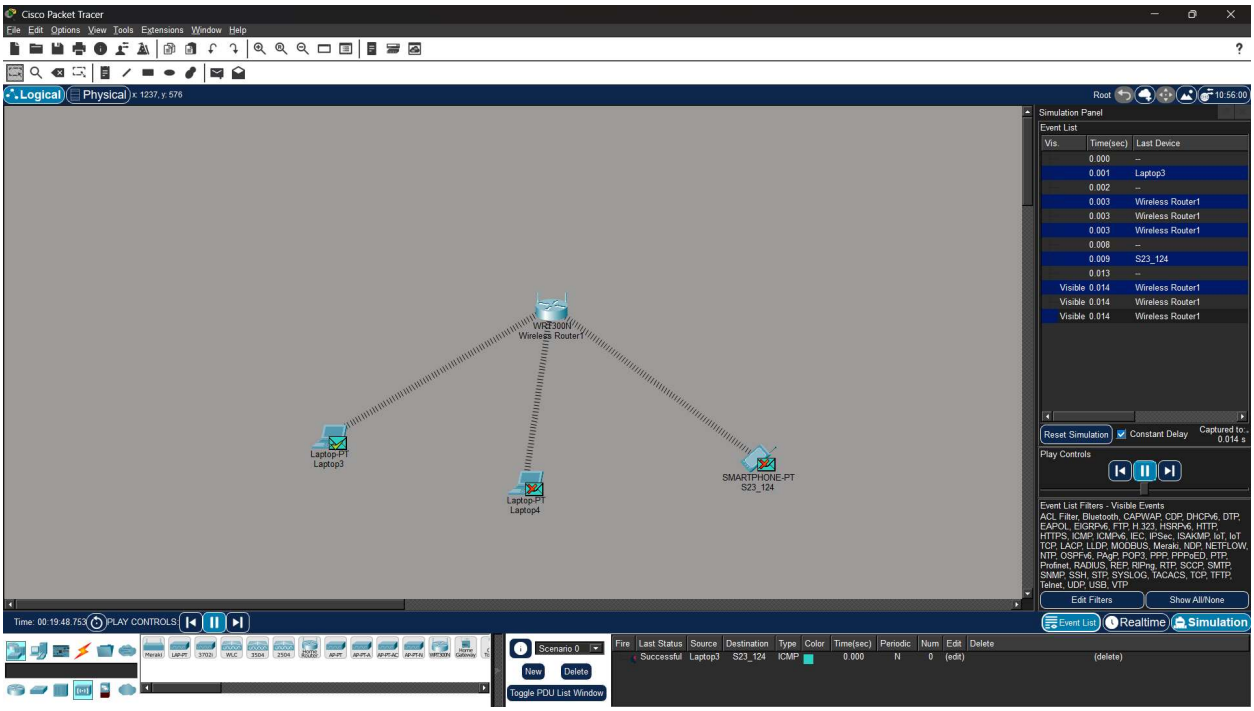
- Unlike the wired experiment, we will not set Static IPs.
- Click **Laptop** > **Desktop** > **IP Configuration**. Ensure **DHCP** is selected.

Device	**SSID**	**Authentication Type**	**Assigned IP Address**
Laptop A	MyWirelessNetwork	WPA2-PSK	192.168.0.102
Laptop B	MyWirelessNetwork	WPA2-PSK	192.168.0.104

Device	SSID	Authentication Type	Assigned IP Address
Smartphone	MyWirelessNetwork	WPA2-PSK	192.168.0.101

Step 6: Testing Signal Transmission

- **Real-time:**
  - Open the Command Prompt on Laptop A and ping the IP address of the Smartphone.
- **Simulation Mode:**
  - Filter for **ICMP**.
  - Use the Add Simple PDU (Envelope) to click Laptop A then Smartphone.
  - Watch the packet travel through the air (wireless media) to the Router and then to the target device.



Observations

- **No Physical Cables:** Communication was established without the use of Copper or Fiber cables, proving the use of Radio Frequency (RF) as the transmission media.

- **SSID Filtering:** Devices could only see and connect to the specific network identified by the SSID.
- **DHCP Automation:** The Wireless Router automatically managed the IP addresses, simplifying the setup of a WLAN.

Conclusion

The experiment successfully demonstrates signal transmission through **Wireless Media**. It proves that a **Wireless Router** acts as an Access Point and a DHCP server, allowing mobile and portable devices to communicate within a defined range using radio waves and security protocols.